

Problem 1

1. [5p] Suppose that we have numbers between 1 and 1000 in a binary search tree, and we want to search for the number **370**. Which of the following sequences **could not be the sequence of nodes examined?** (notice it may be more than one)
- a. 2, 252, 401, 398, 330, 344, 397, 369, 371.
 - b. 924, 220, 902, 244, 898, 258, 362, 370.
 - c. 2, 399, 387, 219, 266, 382, 381, 278, 370.
 - d. 925, 100, 202, 903, 240, 900, 230, 370.
 - e. 935, 278, 347, 621, 299, 392, 358, 369, 370.

Solution : (a), (d) and (e)

Explanation : Let `[low, high]` be the interval that is generated by the process of binary searching.

item (a):

Examined Number	<code>[low, high]</code>
Beginning	<code>[1, 1000]</code>
visit root: 2 < 370 -> right child	<code>[2, 1000]</code>
visit 252 < 370 -> right child	<code>[252, 1000]</code>
visit 401 > 370 -> left child	<code>[252, 401]</code>
visit 398 > 370 -> left child	<code>[252, 398]</code>
visit 330 < 370 -> right child	<code>[330, 398]</code>
visit 344 < 370 -> right child	<code>[344, 398]</code>
visit 397 > 370 -> left child	<code>[344, 397]</code>
visit 369 < 370 -> right child	<code>[369, 397]</code>
visit 371 > 370 -> left child	<code>[369, 371]</code>

- Next, we should visit the left child of 371, which must be 370 since there is only one candidate in interval `[369, 371]`.
- But the sequence did stop at 370, thus this can not be the sequence of nodes examined where 370 is successfully searched.

item (d)

Examined Number	[low, high]
Beginning	[1, 1000]
visit root: 925 > 370 -> left child	[1, 925]
visit 100 < 370 -> right child	[100, 925]
visit 202 < 370 -> right child	[202, 925]
visit 903 > 370 -> left child	[202, 903]
visit 240 < 370 -> right child	[240, 903]
visit 900 > 370 -> left child	[240, 900]
visit 230	⊥

- 230 is not in the interval of BST
- since 230 is in the right subtree of 240, which contracting the fact that the right subtree is greater than the corresponding root.

item (e)

935, 278, 347, 621, 299, 392, 358, 369, 370.

Examined Number	[low, high]
Beginning	[1, 1000]
visit 935 > 370 -> left child	[1, 935]
visit 278 < 370 -> right child	[278, 935]
visit 347 < 370 -> right child	[347, 935]
visit 621 > 370 -> left child	[347, 621]
visit 299	⊥

- 299 is not in the interval of BST
- 299 is in the right subtree of 347, which contracting the fact that the right subtree is greater than the corresponding root.